

Basic Info		
Licensing	Open-source (GPL v3.0)	
Target FPGA or ASIC	VCU128	
Design Language	SystemVerilog	
System-Level Features		
Coupling Type	Tightly-coupled	
Data Types	INT8-32, UINT8-32	
ISA	Custom RISC-V extension	
Peak Ops Per Cycle		
Interfaces	CV-X-IF, AXI	
Supported Operations	The supported operations are addition, subtraction, dot product, cross product, convolution, and shift operations	
Compiler Support	Custom OpenSource RISC-V GCC toolchain	
External Memory Required	Yes	
Cache Or Scratchpad	No	
Architecture		
Type	Von Neumann	
Num Clock Domains	1	
Control Interface	ISAEExtension	Custom
	Memory Mapped Protocol	CV-X-IF
Memory Interface	Interface Protocol	AXI4
	Hierarchy Level	
	Cached	
	IOMMU	
Interrupts	No	
Microarchitecture		
Parametrization	Parametric Num Units	No
	Parametric Config	Yes, number of lanes, internal memory size
Programmability	Programmable Cores	Yes (CVA6)
	ISAEExtension	Custom
Software		
Compiler	Requires Specialized Compiler	True
	Supported Compilers	
Hardware Abstraction Layer	HAL Functions	
highLevelAPI	Has High Level API	No, inline assembly
	SDK Repository	https://github.com/alex2kameboss/MA-riscv-gnu-toolchain
	Has Domain Specific Compiler	
Integration		
IP Distribution	Manifest Type	Bender.yml
	Standalone Simulation	Yes
	Standalone Simulation SW Requirements	QuestaSim
	Integration Docs Or Examples	https://github.com/alex2kameboss/MatrixAcceleratorDemo
Synthesis	Is Synthesizable	Yes
	FPGA Synthesis Examples	Yes
	ASIC Synthesis Examples	
Simulation	Closed Source Simulation	QuestaSim
	Open Source Simulation	
Evaluation	PPA Results Available	Yes